SMOOTHNESS OVER A FIELD IMPLIES GEOMETRIC REGULARITY, GEOMETRIC NORMALITY AND GEOMETRIC REDUCEDNESS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof opens with the author's own parenthetical cross-reference '(See also Lemma ...)'; the parenthetical is reproduced verbatim and carried as a navigation pointer.

Editorial note. The counted spans carry 7 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-smooth-geometrically-normal is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Purely inseparable characterizations of geometric regularity, normality and reducedness. Same chapter, Lemmas lemma-geometrically-regular, lemma-geometrically-normal and lemma-check-only-finite-inseparable-extensions, invoked at proof lines 4627-4630. Supplies the reduction of the three properties to their purely inseparable base changes.
- Smoothness is preserved by base change. Morphisms, Lemma morphisms-lemma-base-change-smooth, invoked at proof lines 4631-4633. Supplies that the base changed morphism is still smooth.
- A scheme smooth over a field is regular, and a regular scheme is normal. Same chapter, Lemma lemma-smooth-regular and Properties, Lemma properties-lemma-regular-normal, invoked at proof lines 4635-4639. Supplies regularity and then normality for a smooth scheme over a field.

Source reference. Original lemma 035Y: sources/stacks-varieties/pinned-original-source.tex, source lines 463–463. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 038O: sources/stacks-varieties/pinned-original-source.tex, source lines 2160–2160. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 038V: sources/stacks-varieties/pinned-original-source.tex, source lines 2424–2424. Pointer outside the counted span, invoked once

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Source reference. Original lemma 038X: sources/stacks-varieties/pinned-original-source.tex, source lines 2544–2544. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 056S: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5736–5736. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01VB: sources/stacks-chow/tags.txt, source lines 2440–2440. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0569: sources/stacks-chow/tags.txt, source lines 6702–6702. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $X \to \mathrm{Spec}(k)$ be a smooth morphism where k is a field. Then X is geometrically regular, geometrically normal, and geometrically reduced over k.*

Proof. (See also Lemma 038X.) Let k' be a finite purely inseparable extension of k. It suffices to prove that $X_{k'}$ is regular, normal, reduced, see Lemmas 038V, 038O, and 035Y. By Morphisms, Lemma 01VB the morphism $X_{k'} \to \mathrm{Spec}(k')$ is smooth too. Hence it suffices to show that a scheme X smooth over a field is regular, normal, and reduced. We see that X is regular by Lemma 056S. Hence Properties, Lemma 0569 guarantees that X is normal. □